

Full 0-D Closed-Loop Fontan Model Technical Reference

Standardized model definition, equations, segments, and free parameters

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1 Full 0-D Closed-Loop Fontan Model Technical Reference

This document is generated from repository sources by `scripts/docs/build_model_reference_pdfs.py`. Edit the model config, implementation notes, schematic, or this generator, then regenerate the markdown and PDF together.

1.1 Model Construction

1.1.1 Scope and Status

Accepted full 0-D reference model.

This model is the accepted closed-loop 0-D reference circulation.

It uses a PhysioBlocks active spherical single ventricle, an active time-varying-elasticity atrium, R-L valves, lumped vascular compliances, systemic resistive beds, Fontan conduit elements, pulmonary RCR beds, and an optional fenestration shunt.

No inlet pressure, outlet pressure, or prescribed inflow boundary condition drives the baseline circulation.

1.1.2 Schematic

1.1.3 Accepted Components

- active atrium and active spherical ventricle

Closed-loop zero-dimensional Fontan schematic

Active atrium, expanded aortic tree, systemic vascular beds, TCPC conduits, and pulmonary Windkessel beds. Nominal flow follows the arrows.
Task 004 calibrated baseline: HR 60-80 bpm, 25 s settling, scaled heterosystemic/pulmonary/TCPC parameters; topology unchanged.

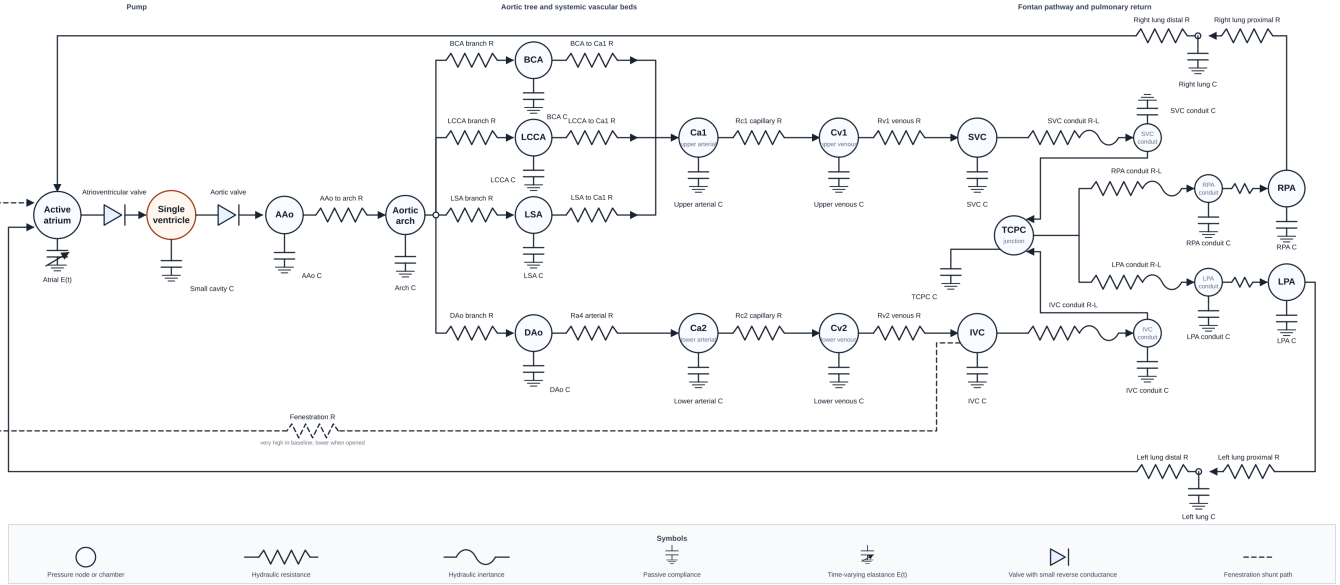


Figure 1: full_0d schematic

- atrioventricular and aortic valve R-L blocks
- ascending aorta, aortic arch, BCA, LCCA, LSA, and DAo tree
- upper and lower systemic vascular beds
- SVC, IVC, RPA, and LPA Fontan conduit states
- right and left pulmonary RCR Windkessel beds
- high-resistance baseline fenestration path

1.1.4 Authoritative Baseline Config

The executable topology and free-parameter values are taken from `models/full_0d/configs/fontan_0d_baseline.jsonc`.

- pressure nodes: 21
- blocks/segments: 48
- free parameter entries: 91
- boundary conditions: 0

1.1.5 Scenario Configs

- `models/full_0d/configs/fontan_0d_baseline.jsonc`
- `models/full_0d/configs/fontan_0d_fenestration.jsonc`
- `models/full_0d/configs/fontan_0d_lpa_obstruction.jsonc`
- `models/full_0d/configs/fontan_0d_smoke.jsonc`
- `models/full_0d/configs/fontan_0d_vasodilation.jsonc`

1.1.6 Pressure Nodes

- atrial
- cavity
- aao
- aortic_arch
- bca
- lcca
- lsa
- upper_art
- upper_ven

- dao
- lower_art
- lower_ven
- svc
- svc_conduit
- ivc
- ivc_conduit
- tcpc
- rpa_conduit
- lpa_conduit
- rpa
- lpa

1.1.7 Block Type Counts

Block type	Count
c_block	20
rc_block	18
rcr_block	2
spherical_cavity_block	1
time_varying_elastance_atrium_block	1
valve_rl_block	6

1.2 Governing Equations

The sign convention follows PhysioBlocks local block fluxes. For a two-node flow element, local node 1 is the first node listed in the config and local node 2 is the second node listed in the config.

1.2.1 Nodal Conservation

At every pressure node, the algebraic/differential network residual is the sum of all block flux contributions attached to that node:

$$\sum_{b \in \mathcal{B}(i)} Q_{b,i} = 0.$$

Storage blocks contribute pressure derivatives to this same residual, so closed-loop volume conservation is enforced through the connected block equations rather than through prescribed boundary flow.

1.2.2 Passive Compliance Block

For a c_block at pressure node P with capacitance C:

$$Q = -C \frac{dP}{dt}.$$

The saved stored volume is proportional to pressure in the local linear compliance approximation:

$$V = CP.$$

1.2.3 Pure RC Resistor Convention

This repository uses rc_block with zero capacitance as a pure resistive link. PhysioBlocks defines the local fluxes as:

$$Q_1 = \frac{P_2 - P_1}{R},$$

$$Q_2 = \frac{P_1 - P_2}{R} - C \frac{dP_2}{dt}.$$

When $C = 0$, the block is used as a pure resistor. To represent an upstream-to-downstream path with positive physical flow from upstream to downstream, the configs assign local node 2 to the upstream pressure and local node 1 to the downstream pressure.

1.2.4 Hydraulic R-L Link

The local quasi-vessel R-L element uses positive internal flow from local node 1 to local node 2:

$$L \frac{dQ}{dt} + RQ = P_1 - P_2,$$

$$Q_1 = -Q, \quad Q_2 = Q.$$

This is the repeated segment equation used by the quasi 0-D/1-D chains.

1.2.5 Valve R-L Block

The R-L valve block has local positive flow from node 1 to node 2 and switches conductance according to flow direction:

$$L \frac{dQ}{dt} + P_2 - P_1 + R(Q)Q = 0,$$

$$R(Q) = \begin{cases} 1/G_f, & Q > 0, \\ 1/G_b, & Q < 0, \end{cases}$$

$$Q_1 = -Q, \quad Q_2 = Q.$$

G_f is the forward conductance and G_b is the backward conductance.

1.2.6 Pulmonary RCR Windkessel

For an `rcr_block` with inlet pressure P_1 , outlet pressure P_2 , middle pressure P_m , proximal resistance R_1 , distal resistance R_2 , and compliance C :

$$Q_1 = \frac{P_m - P_1}{R_1},$$

$$Q_2 = \frac{P_m - P_2}{R_2},$$

$$\frac{P_1 - P_m}{R_1} + \frac{P_2 - P_m}{R_2} - C \frac{dP_m}{dt} = 0.$$

1.2.7 Active Atrium

The active atrium is a one-node time-varying elastance chamber:

$$E(t) = E_{min} + (E_{max} - E_{min})a(t),$$

$$V_a(t) = V_{0,a} + \frac{P_a(t) - P_{ext}}{E(t)},$$

$$Q_a = -\frac{dV_a}{dt}.$$

The activation $\mathbf{a}(\mathbf{t})$ is a raised-cosine pulse over the configured start, peak, and end phase of the cardiac cycle.

1.2.8 Spherical Ventricular Cavity

The active ventricular cavity stores volume through the spherical cavity displacement y , reference radius R_0 , and wall thickness d_0 :

$$V(y) = \frac{4\pi}{3} \left[R_0 \left(1 + \frac{y}{R_0} \right) - \frac{d_0}{2 \left(1 + \frac{y}{R_0} \right)^2} \right]^3,$$

$$Q_v = -\frac{dV(y)}{dt}.$$

The pressure-displacement relation comes from the configured PhysioBlocks spherical dynamics, velocity law, passive rheology, and active macro-Huxley submodels. The corresponding free parameters are listed in the parameter inventory below with the `cavity.*` prefix.

1.3 Segment Inventory

Each row is a block or segment in the accepted baseline config. The parameter column lists explicit block fields and all config parameters sharing the block-name prefix.

Segment/block	Type	Local nodes	Free-parameter fields
<code>cavity</code>	<code>spherical_cavity_block</code>	1: cavity	<code>disp =</code> <code>cavity.dynamics.disp;</code> <code>radius -> heart_radius;</code> <code>thickness -></code> <code>heart_thickness;</code> <code>cav-</code> <code>ity.dynamics.damping_coef;</code> <code>cav-</code> <code>ity.dynamics.hyperelastic_cst;</code> <code>cavity.dynamics.vol_mass;</code> <code>cav-</code> <code>ity.rheology.active_law.activation;</code> <code>cav-</code> <code>ity.rheology.active_law.crossbridge_sti</code> <code>cav-</code> <code>ity.rheology.active_law.destruction_ra</code> <code>cav-</code> <code>ity.rheology.active_law.starling_abscis</code> <code>cav-</code> <code>ity.rheology.active_law.starling_ordin</code> <code>cav-</code> <code>ity.rheology.damping_parallel;</code> <code>cav-</code> <code>ity.rheology.series_stiffness;</code> <code>cav-</code> <code>ity.velocity_law.scheme_ts_hht</code> <code>backward_conductance -></code> <code>valves.backward_conductance;</code> <code>valve_atrium.conductance;</code> <code>valve_atrium.inductance;</code> <code>valve_atrium.scheme_ts_flux</code>
<code>valve_atrium</code>	<code>valve_rl_block</code>	1: atrial; 2: cavity	

Segment/block	Type	Local nodes	Free-parameter fields
valve_arterial	valve_rl_block	1: cavity; 2: aao	backward_conductance -> valves.backward_conductance; valve_arterial.conductance; valve_arterial.inductance; valve_arterial.scheme_ts_flux
capacitance_valve	c_block	1: cavity	capacitance_valve.capacitance
active_atrium	time_varying_elastance_atrium_block	1: cavity	pressure_external -> pleural.pressure; elastance_min -> ac- tive_atrium.elastance_min; elastance_max -> ac- tive_atrium.elastance_max; unstressed_volume -> ac- tive_atrium.unstressed_volume; activation_start -> ac- tive_atrium.activation_start; activation_peak -> ac- tive_atrium.activation_peak; activation_end -> ac- tive_atrium.activation_end; heartbeat_duration -> heartbeat_duration
aao_compliance	c_block	1: aao	aao_compliance.capacitance
aortic_arch_compliance	c_block	1: aortic_arch	aortic_arch_compliance.capacitance
bca_compliance	c_block	1: bca	bca_compliance.capacitance
lcca_compliance	c_block	1: lcca	lcca_compliance.capacitance
lsa_compliance	c_block	1: lsa	lsa_compliance.capacitance
upper_ca1	c_block	1: upper_art	upper_ca1.capacitance
upper_cv1	c_block	1: upper_ven	upper_cv1.capacitance
dao_compliance	c_block	1: dao	dao_compliance.capacitance
lower_ca2	c_block	1: lower_art	lower_ca2.capacitance
lower_cv2	c_block	1: lower_ven	lower_cv2.capacitance
svc_compliance	c_block	1: svc	svc_compliance.capacitance
ivc_compliance	c_block	1: ivc	ivc_compliance.capacitance
tcpc_compliance	c_block	1: tcpc	tcpc_compliance.capacitance
svc_conduit_compliance	c_block	1: svc_conduit	svc_conduit_compliance.capacitance
ivc_conduit_compliance	c_block	1: ivc_conduit	ivc_conduit_compliance.capacitance
rpa_conduit_compliance	c_block	1: rpa_conduit	rpa_conduit_compliance.capacitance
lpa_conduit_compliance	c_block	1: lpa_conduit	lpa_conduit_compliance.capacitance
rpa_compliance	c_block	1: rpa	rpa_compliance.capacitance
lpa_compliance	c_block	1: lpa	lpa_compliance.capacitance
svc_conduit_rl	valve_rl_block	1: svc; 2: svc_conduit	conductance -> svc_conduit_rl.conductance; backward_conductance -> svc_conduit_rl.backward_conductance; scheme_ts_flux -> conduit.scheme_ts_flux; svc_conduit_rl.inductance
svc_conduit_junction	rc_block	1: tcpc; 2: svc_conduit	capacitance -> zero_capacitance; svc_conduit_junction.resistance
aao_arch	rc_block	1: aortic_arch; 2: aao	capacitance -> zero_capacitance; aao_arch.resistance
arch_bca	rc_block	1: bca; 2: aortic_arch	capacitance -> zero_capacitance; arch_bca.resistance

Segment/block	Type	Local nodes	Free-parameter fields
upper_bca_to_ca1	rc_block	1: upper_art; 2: bca	capacitance -> zero_capacitance; up- per_bca_to_ca1.resistance
arch_lcca	rc_block	1: lcca; 2: aortic_arch	capacitance -> zero_capacitance; arch_lcca.resistance
upper_lcca_to_ca1	rc_block	1: upper_art; 2: lcca	capacitance -> zero_capacitance; up- per_lcca_to_ca1.resistance
arch_lsa	rc_block	1: lsa; 2: aortic_arch	capacitance -> zero_capacitance; arch_lsa.resistance
upper_lsa_to_ca1	rc_block	1: upper_art; 2: lsa	capacitance -> zero_capacitance; up- per_lsa_to_ca1.resistance
upper_rc1	rc_block	1: upper_ven; 2: upper_art	capacitance -> zero_capacitance; upper_rc1.resistance
upper_rv1	rc_block	1: svc; 2: upper_ven	capacitance -> zero_capacitance; upper_rv1.resistance
arch_dao	rc_block	1: dao; 2: aortic_arch	capacitance -> zero_capacitance; arch_dao.resistance
lower_ra4	rc_block	1: lower_art; 2: dao	capacitance -> zero_capacitance; lower_ra4.resistance
lower_rc2	rc_block	1: lower_ven; 2: lower_art	capacitance -> zero_capacitance; lower_rc2.resistance
lower_rv2	rc_block	1: ivc; 2: lower_ven	capacitance -> zero_capacitance; lower_rv2.resistance
ivc_conduit_rl	valve_rl_block	1: ivc; 2: ivc_conduit	conductance -> ivc_conduit_rl.conductance; backward_conductance -> ivc_conduit_rl.backward_conductance; scheme_ts_flux -> conduit.scheme_ts_flux; ivc_conduit_rl.inductance
ivc_conduit_junction	rc_block	1: tcpc; 2: ivc_conduit	capacitance -> zero_capacitance; ivc_conduit_junction.resistance
rpa_conduit_rl	valve_rl_block	1: tcpc; 2: rpa_conduit	conductance -> rpa_conduit_rl.conductance; backward_conductance -> rpa_conduit_rl.backward_conductance; scheme_ts_flux -> conduit.scheme_ts_flux; rpa_conduit_rl.inductance
rpa_conduit_out	rc_block	1: rpa; 2: rpa_conduit	capacitance -> zero_capacitance; rpa_conduit_out.resistance

Segment/block	Type	Local nodes	Free-parameter fields
lpa_conduit_rl	valve_rl_block	1: tepc; 2: lpa_conduit	conductance -> lpa_conduit_rl.conductance; backward_conductance -> lpa_conduit_rl.backward_conductance; scheme_ts_flux -> conduit.scheme_ts_flux; lpa_conduit_rl.inductance
lpa_conduit_out	rc_block	1: lpa; 2: lpa_conduit	capacitance -> zero_capacitance; lpa_conduit_out.resistance
right_lung	rcr_block	1: rpa; 2: atrial	pressure_mid = right_lung.pressure_mid; resistance_1 -> right_lung.resistance_1; resistance_2 -> right_lung.resistance_2; capacitance -> right_lung.capacitance
left_lung	rcr_block	1: lpa; 2: atrial	pressure_mid = left_lung.pressure_mid; resistance_1 -> left_lung.resistance_1; resistance_2 -> left_lung.resistance_2; capacitance -> left_lung.capacitance
fenestration	rc_block	1: atrial; 2: ivc	capacitance -> zero_capacitance; fenestration.resistance

1.4 Free Parameters

Unless a parameter states otherwise in the config, units follow the repository SI convention: pressure in Pa, flow in $\text{m}^3 \text{s}^{-1}$, resistance in Pa s m^{-3} , capacitance in $\text{m}^3 \text{Pa}^{-1}$, inertance in $\text{Pa s}^2 \text{m}^{-3}$, volume in m^3 , length in m, and time in s.

The entries below are the complete `parameters` dictionary from the authoritative baseline config. Derived-expression entries are shown as compact JSON.

- `aao_arch.resistance = 2133152`
- `aao_compliance.capacitance = 4.0003400289e-09`
- `active_atrium.activation_end = 0.98`
- `active_atrium.activation_peak = 0.9`
- `active_atrium.activation_start = 0.78`
- `active_atrium.elastance_max = 66661000`
- `active_atrium.elastance_min = 16665250`
- `active_atrium.unstressed_volume = 4e-05`
- `active_law.activation.max = 35`
- `active_law.activation.min = -20`
- `aortic_arch_compliance.capacitance = 5.00042503613e-09`
- `arch_bca.resistance = 5332880`
- `arch_dao.resistance = 19198368`
- `arch_lcca.resistance = 7999320`
- `arch_lsa.resistance = 7999320`
- `bca_compliance.capacitance = 3.00025502168e-09`
- `capacitance_valve.capacitance = 5e-12`
- `cavity.dynamics.damping_coef = 70`

- cavity.dynamics.hyperelastic_cst = [444.0,2.9,69.0,6.5]
- cavity.dynamics.vol_mass = 1000
- cavity.rheology.active_law.activation = {"alpha": "diastole_scaling_factor", "phases": [0,0,1,1,1,0,0], "ref"
- cavity.rheology.active_law.crossbridge_stiffness = 273000
- cavity.rheology.active_law.destruction_rate = 12
- cavity.rheology.active_law.starling_abscissas = [-0.1668,-0.0073,0.0534,0.0969,0.1326,0.2016,0.4663,0.91
- cavity.rheology.active_law.starling_ordinates = [0.0,0.5614,0.7748,0.8933,0.9618,1.0,1.0,0.1075,0.0]
- cavity.rheology.damping_parallel = 70
- cavity.rheology.series_stiffness = 100000000
- cavity.velocity_law.scheme_ts_hht = 0.4
- conduit.scheme_ts_flux = 0.25
- dao_compliance.capacitance = 9.00076506503e-09
- diastole_scaling_factor = 0.8
- fenestration.resistance = 1.33322e+14
- heart_contractility = 37800
- heart_radius = 0.02431
- heart_rate = 69.9300699301
- heart_thickness = 0.0080795
- heartbeat_duration = {"factors": [60.0], "inverses": ["heart_rate"], "type": "product"}
- ivc_compliance.capacitance = 2.25019126626e-07
- ivc_conduit_compliance.capacitance = 1.20010200867e-09
- ivc_conduit_junction.resistance = 219981.3
- ivc_conduit_rl.backward_conductance = 2.39254786418e-07
- ivc_conduit_rl.conductance = 2.39254786418e-07
- ivc_conduit_rl.inductance = 80000
- lcca_compliance.capacitance = 1.50012751084e-09
- left_lung.capacitance = 3.00025502168e-08
- left_lung.resistance_1 = 8319292.8
- left_lung.resistance_2 = 12478939.2
- lower_ca2.capacitance = 2.50021251806e-08
- lower_cv2.capacitance = 9.00076506503e-08
- lower_ra4.resistance = 43196328
- lower_rc2.resistance = 140388066
- lower_rv2.resistance = 32397246
- lpa_compliance.capacitance = 1.12509563313e-08
- lpa_conduit_compliance.capacitance = 1.20010200867e-09
- lpa_conduit_out.resistance = 173318.6
- lpa_conduit_rl.backward_conductance = 3.03669536607e-07
- lpa_conduit_rl.conductance = 3.03669536607e-07
- lpa_conduit_rl.inductance = 60000
- lsa_compliance.capacitance = 1.50012751084e-09
- pleural.pressure = 0
- right_lung.capacitance = 3.00025502168e-08
- right_lung.resistance_1 = 5759510.4
- right_lung.resistance_2 = 8639265.6
- rpa_compliance.capacitance = 1.12509563313e-08
- rpa_conduit_compliance.capacitance = 1.20010200867e-09
- rpa_conduit_out.resistance = 119989.8
- rpa_conduit_rl.backward_conductance = 4.38633775099e-07
- rpa_conduit_rl.conductance = 4.38633775099e-07
- rpa_conduit_rl.inductance = 60000
- svc_compliance.capacitance = 1.50012751084e-07
- svc_conduit_compliance.capacitance = 1.20010200867e-09
- svc_conduit_junction.resistance = 293308.4
- svc_conduit_rl.backward_conductance = 1.79441089813e-07
- svc_conduit_rl.conductance = 1.79441089813e-07
- svc_conduit_rl.inductance = 80000
- tcpc_compliance.capacitance = 1.20010200867e-09

- `upper_bca_to_ca1.resistance = 49595784`
- `upper_ca1.capacitance = 1.50012751084e-08`
- `upper_cv1.capacitance = 6.00051004335e-08`
- `upper_lcca_to_ca1.resistance = 99458212`
- `upper_lsa_to_ca1.resistance = 99458212`
- `upper_rc1.resistance = 156439020.571`
- `upper_rv1.resistance = 67045294.5306`
- `valve_arterial.conductance = 1.3e-05`
- `valve_arterial.inductance = 30000`
- `valve_arterial.scheme_ts_flux = 0.25`
- `valve_atrium.conductance = 9e-06`
- `valve_atrium.inductance = 1000`
- `valve_atrium.scheme_ts_flux = 0.25`
- `valves.backward_conductance = 5e-12`
- `zero_capacitance = 0`

1.5 Documentation and Regeneration

Model-local documentation artifacts:

- `models/full_0d/README.md`
- `models/full_0d/docs/full_0d_schematic.svg`
- `models/full_0d/docs/full_0d_schematic.png`
- `models/full_0d/docs/implementation_notes.md`
- `models/full_0d/docs/full_0d_technical_reference.md`
- `models/full_0d/docs/full_0d_technical_reference.pdf`

Regenerate the technical reference source and PDF with:

```
python3 scripts/docs/build_model_reference_pdfs.py --model full_0d
```

1.6 Current Limitations

- The model is a calibrated computational development artifact, not a clinically validated patient-specific simulator.
- The aortic and Fontan pathways are lumped 0-D approximations rather than true spatially resolved 1-D domains.
- Scenario files inherit the baseline calibration and are validation cases, not independently retuned models.

The model parameters and standardized data are for computational development and calibration workflows. Simulation outputs must not be presented as clinically validated without separate validation and documentation.